

SI409: Why Trend Following Is Changing Faster Than Ever ft. Yoav Git

Podcast: Top Traders Unplugged

AI Executive Summary

Core Thesis The episode explores the evolving landscape of systematic investing, focusing on how shifts in market microstructures, new financial instruments, and changing investor behaviors are reshaping risk and opportunity for trend followers. The central argument is that while traditional index-based trend following faces headwinds from increased mean reversion (driven by 0DTE options) and rising idiosyncratic risk, significant alpha opportunities now exist in deeper layers of the market, including industry sectors, thematic narratives, single-stock futures, and specialized commodity curves.

Key Arguments & Data

- * **New Market Instruments:** * The CME is launching single-stock futures for approximately 50 leading US stocks, which simplifies shorting by removing borrowing/lending complexities.
- * Kalshi is introducing forward curves for compute power (e.g., Nvidia chips), effectively creating a new class of "commodity" trading based on technological utility.
- * **Market Structure & Volatility Dynamics:**
- * **The Options Explosion:** The rise of Zero Day to Expiry (0DTE) options, largely driven by retail investors selling at-the-money straddles, has created concentrated gamma for market makers. This force acts as a "counter-trend" mechanism, pulling prices back toward the strike and increasing index mean reversion.
- * **High Dispersion:** The dispersion index is at an all-time high, meaning individual stock volatility is decoupling from index volatility; idiosyncratic risk in equities is currently much higher than systematic index risk.
- * **Tick Size Impact:** As equity prices rise, the "volatility-normalized tick size" decreases, making intraday/short-term trend following more difficult due to changes in order book sparsity and microstructure.
- * **Risk Profiles & Blow-up Risks:** * Data from Sins Capital Macro Terminal ranks hedge fund strategies by liquidation risk: Short Volatility and Short Gamma carry the highest "blow-up" likelihood, while CTA managed futures and diversified macro funds are among the lowest risk.
- * The extreme volatility of single stocks (e.g., IBM's 25% intraday drop) serves as a warning against over-leveraging in concentrated equity positions.
- * **Thematic & Narrative Investing:**
- * **The "Missing" Variance:** Since index and industry factors explain only about 50% of

volatility, "themes" (e.g., the AI boom, geopolitical shifts) represent the remaining unexplained variance. * **Narrative Momentum:** Research shows that investors often underreact to news-driven narratives initially, creating a window for trend following within specific themes before the market fully prices in the event. * **Thematic Risk:** Using broad ETFs to capture themes can be risky because they are often over-exposed to standard industry or country factors; bespoke construction is preferred for "pure" exposure.

Actionable Takeaways

- * **Expand Diversification Horizons:** Investors should move beyond simple index futures. To find uncorrelated alpha, look toward industry-specific sectors, factor-based exposures (e.g., growth/value), and the increasingly volatile single-stock universe.
- * **Monitor Dispersion & Idiosyncratic Risk:** When trading equities, recognize that the index is currently behaving in a mean-reverting, low-volatility manner due to options hedging; look for opportunities where individual stocks are diverging from the aggregate trend.
- * **Utilize "Thematic" Research:** Incorporate news-based and narrative-driven analysis into systematic models. Identifying themes (like semiconductors or energy transitions) before they become mainstream "industry" trends can capture the underreaction phase of a move.
- * **Apply Caution to Short Volatility Strategies:** Given the high risk of forced liquidation in short gamma/short vol strategies, ensure that any volatility-selling component is strictly size-controlled and part of a broader, much more defensive CTA framework.

Full Transcript

Welcome to Top Traders Unplugged. In markets, success doesn't come from predicting what happens next. It comes from being prepared for what you can't predict. In each episode, we go deep with some of the world's most thoughtful minds in investing, economics, and beyond to understand how they think, how they prepare, and how they decide, and the experiences that shaped how they see the world. No noise, no shortcuts, just real conversations to help you think better and invest with confidence. Welcome or welcome back to this week's edition of the Systematic Investors Series where you have Git and I, Niels Kastrup last week where each week we take the pulse of the global market through the lens of a rules-based investor. It is wonderful to have you back this week. I hope you're doing well despite the challenging result in the football last night.

I've got to say, as an English supporter, we're kind of used to that behaviour. It's very much trend following. Yeah, that is true. Unfortunately, it's one of those trends you don't really want to follow nor set. No, absolutely. You become very emotionally involved and invested throughout the tournament, only for counter-trend to disappoint you right at the end. Yeah, well, at least we now have a final with a European team and a South American team, so we get a little bit of diversification in terms of... style. Absolutely, diversification is very important. Good stuff. All right. We've got a pretty full plate today. Great line of a topic. Thank you so much for bringing them along today. We'll be diving into all of that in a few minutes. But as usual, I'm always curious to find out what's been on your radar the last few weeks or a few days for that matter.

Thank you. So the thing that has been on my radar are actually new products coming to market. So we've seen the CME has announced that they are going to launch 50-odd leading US stocks as the single stock futures on the CME, which I think is really, really exciting. And at the same time, Kalshi is announcing forward curves for compute power for the cost of compute power. So that's like a different a different type of commodity which we've never seen before which is like using various chips of Nvidia. And that's really exciting to me. But I think it brings together one aspect

that people kind of normally, as a mathematician, you kind of ignore it. But when you're a practitioner, one of the things that is really important is actually thinking about the tax implications or various practical implications of trading it. So I've spent the last week figuring out what would be the tax implication of trading single-stick futures.

Because the treatment of the dividend, the treatment of the price rise, that's actually quite important. It's very clear for like index futures, everybody knows what it is. For single stocks, less so. And for Kalshi, it's even more complicated. So the tax status, whatever it is, it's still unclear. And I think if we want to see growth and innovation, it's the practical things which... often stopping you from getting involved in the markets, which where you kind of want to. Yeah. I mean, single stock futures, there is definitely a little bit of a debate among CTA practitioners for sure. And I don't know that that many of our colleagues have embraced them. I do know that some have embraced them and feel very strongly about the advantage of trading single stock futures rather than just trading the indices.

But at least it gives more choice, I guess. But I will be interested in seeing what the adoption rate, at least among what I consider kind of the CTA world, I'm sure it'll be adopted by the wider hedge fund audience. But yeah, very exciting. And there seems to be a lot of innovation going on in terms of of futures and markets almost to a point where you think well do we really need another one of these but maybe we do. It's not necessarily just for CTAs but just the ability to short. stock, right? It's not something that it's very easy for, it's very easy for a retail investor, right? But if you have a future contract, then it suddenly becomes possible because you don't have to worry about shorting and borrowing and that aspect of it. So I think actually I'm kind of, I think that will be very attractive.

even not for CTAs, but I think it will be attractive to CTAs as well, but also for the audience in general, the ability to short specific stocks will be useful. Yeah, I mean, it's usually a good idea to give people more choice, although I will say, and that's not the topic for our discussion today, so I'm not inviting to it for a big debate here, but I will say with some of the new stuff that's coming out and where you give, especially you have the focus on retail, And I'm just thinking, hmm, is this really a good idea to give these

financial instruments to people who do not do this as their profession? I'm not so sure. And it's not because, oh, let's just keep it for the professionals and so that we can make more money. That's really not what I mean. But being short of stock.

can be lethal because it's unlimited upside, sorry, unlimited downside if you're short. And there's a takeover or whatever that might be. So all I'm just saying is, yeah, great. No, 100%. I mean, I think it's a really interesting discussion, by the way, about the retail investors. And I think we'll come to it when we look at certain observations that CFM has made. But from our perspective, We look for diversity and we look for new risk factors and compute powers in AI is something which is very much of interest and that's why I think both CME and Culture are looking at it and single stocks is again potentially a way to diversify your portfolio and it's interesting to us as a result of that.

I'll try and tie in some of the things you've mentioned in my observations, although it may fail spectacularly, but when it comes to kind of energy and compute power and all that, on my radar this week was a very energetic and powerful experience which was, and there might be a lot of people who won't share my taste in music, but I did go this week to watch the the tour of Bruno Mars live in Milan. And all I can say is that if people have not experienced him live, it was quite an experience. And the energy level for two hours straight was amazing even through the fact that it was 35 degrees Celsius while he was doing all of that. But it was an amazing evening and of course special to share it with you.

with your family for sure. The next thing I wanted to mention, not entirely sure where I saw this could have been on Twitter, but there was a, I think it's a company called Sins Capital Macro Terminal. It was a, it was a, what do you call it, like a, not a meme, but a, anyway, one of these illustrations they had where they had put in rank, different hedge fund strategies ranked by blow up. forced liquidation risk. I don't know if you saw that, but it's quite interesting. So the worst one, the nuclear one, was short vol, tail risk premia short gamma. That's kind of the one they classify as having the highest blow-up likelihood. Then we have a lot of other things. But then, of course, my eye went straight to looking for CTA managed futures.

CTA Ministry just came in as number 15 out of 17 with a green collar, lower, it says, it said, yeah, so really on the lower side of blow up risk, which I completely agree with. It's completely understandable, yeah. So Vault targeted diversified macro was the second lowest and then long only hedge fund like equity, whatever that means. I wonder if they meant private equity. I think private equity, I can't remember exactly, but certainly from the way people operate, long-only equity, less likely to blow, CTA, anybody who manages the risk, so CTAs and volatility driven macros, is just. We are more defensive. We are likely to have upside rather than downside. So the blow-up is better. I mean, it doesn't mean that when you have a position and the world blow-ups, you are immediately protected, right? We are like second responder. But certainly being short of all is kind of dangerous, especially when the world blows up. Yeah. I mean, long-only hedge fund. I mean, for those who are old enough to remember, I mean, if you're a long-only Nasdaq in 2000 and...

One, you would have been down 85% at some point. So to say that nothing can happen by being long only even in an index, it might be stretching a little bit. Speaking of risk, couldn't help notice that IBM lost 69 billion of market value in one day. And that's about 25% in one day. Looking at the price of that stock, it traded. on 2nd of June only about six weeks ago at 332 that's an all-time high as far as I'm aware and earlier this week we were down to 213 that's one third of the value in only about six weeks which again goes to my point maybe we don't need to give people too many instruments where they can add additional leverage or risk when you see what happens just with individual stocks outright in terms of How much they can really move in the short space of time? I mean, 25% of the. Yeah, that's a big move. I think there are lots of questions about AI at the moment and about the CapEx, actually the capital expenditure. And the reason for that is really we're spending quite a lot of money on chips and Nvidia is doing well out of it.

But these chips don't last forever. In the boom and bust you mentioned in 2001, there was a lot of boom, but the cables that were laid in that time, on that money, on that capex, have served us very well for the next 20 years, 25 years. But all this money which is being thrown at AI is thrown at chips that will last maybe for two, three years, and then they become old and

obsolete. So there's a lot of cupcakes, which if it does go wrong and AI doesn't materialize itself, basically, there's going to be zero recovery. You're not getting anything out of it. On that note, I actually saw an interview that was posted on X with Jensen Wong from NVIDIA. And I will say, not that I haven't watched much of his stuff, of his time of familiar with him, but he was very impressive. Actually, he very well-argued. He argued very well for...

why he didn't think necessarily the AI meant that everybody's going to lose their job. But he did say you might lose your job to someone using AI if you don't educate yourself and get to know about it. And he might be right on that. But anyways, we've got an interesting conversation he did. And I think it was that one of these venture capital companies could have been Andreessen Horowitz in Silicon Valley where he was invited to do like a fireside. 100% because people are not necessarily hiring juniors, but they hire juniors because they know how to use it. Yes, exactly. Ask all these, we need to adapt to survive. Yes, no, I completely agree. We also need to adapt in the world of trend following and July is another month of a little bit of adaptation as far as I can tell.

It's somewhat stable, but I did see some managers with a little bit more downside pressure than elsewhere. And of course, it might feel a little bit more stable than June, but there's still a lot of noise coming from the geopolitical scene. And so we do see a little bit of a negative bias overall. I mean, oil risk, of course, that's back now with the ceasefire no longer being in place and all the headlines. Yeah, I mean, it might be interesting, let me put it like this, to see whether our space got a little bit divided in terms of whether we got caught or who got caught short in energies when they pulled off. And then we got this sudden move back up again. So I think time frame or speed as we like to refer to may have had a little bit to say this month more than usual. But also, of course, we've got tech stocks selling off in the US. Some of the Japanese markets are selling off this month as well. And we also see a little bit of a directional change in things like Canadian dollar, as far as I could tell. So yeah, what's your observation so far? No, I think July has been countertrend.

In general both you know in fixed income certainly and an FX as you mentioned There's some idiosyncratic risk in Japanese bonds. There's a question of repatriation of FX back to Japan. But also, I think in commodity, you're making exactly the right observation. And it was really much on the cusp. Some people would have been short oil already and would have been caught out. And some people would have been still, if you're a very slow trader, you would have been still potentially long and then benefited from this. So I think that we're going to get a little bit of dispersion in commodities this month. So it continues to be an interesting period as it always is. My trend barometer kind of reflects that. It ended yesterday at 45. That's a completely neutral reading and I would say it's fairly much in line with what we see in terms of the performance numbers. So these numbers are as of Tuesday evening, as they only update later on today for Wednesday. But beta 50 down three basis points are pretty flat. Still up eight and a half for the year. Sock Gen CD index down 38 basis points in July. Still up about 9% for the year. Sock Gen trend down 33 basis points up about 8.76% for the year. And the short term traders index, interestingly enough, as we will We're going to be talking about short term a little later. And that's hurting a little bit more down 1.12% in July, still up 4% for the year, but that is a lower vol index. So losing more than the trend index means that it's probably under a little bit more pressure this month. MSCI World in the traditional world is up 1.1% so far in July as of last night, up 11.15%.

We have the S&P US aggregate bond index that's down 43 basis points, but still up 41 basis points for the year. And then the S&P 500 up 1% for the month of July up 11.32% of doing pretty well overall. All right. So Marcus wrote in a couple of weeks ago, but we did not get to it until today. And Marcus Wright, thank you for the fantastic podcast. I've been listening for about three years and I have learned a tremendous amount. So we appreciate that feedback, of course. My question specifically concerns systematic trend following in non-financial futures where different contract months can represent meaningfully different economic exposures due to seasonality, storage, carry and delivery dynamics. Most trend following implementations appear to generate signals from a continuous front month future series and then preserve exposure mechanically through contract roles. However, I've never heard the more fundamental model in question

discussed. What return process are we actually trying to forecast? Are we trying to model the underlying spot process, the return process of the currently tradable futures contract or the return process of a continuously rolled investment strategy?

In markets such as net gas and agricultural commodities, neighboring expiries can have materially different behavior and even different risk characteristics. If that's true, what is the theoretical or empirical justification for treating the continuous fund month series as the main object for signal generation? Has anyone tested generating signals solely on the history of the currently selected tradeable contract? and revalidating positions when changing contracts rather than inherently inheriting exposure mechanically through the role. I'm not asking about proprietary implementation details. I'm generally interested in the modeling philosophy behind systematic trend following and whether this question has been explored in the literature or by practitioners. Many thanks for everything you do. And I hope you find the question interesting. Actually, I do Marcus. So, and I have the perfect guess to answer it for you. I hope.

And therefore you're up over to you. I'll get to say that was such a detailed and actually knowledgeable question. Like Marcus literally answered the question himself. But he's not going to be surprised to know that we have looked at all of these. within our company we have a commodity fund that actually does trade the full set of sort of the full curve so to speak and I think the question actually makes a lot of sense because unlike normal assets which stay which persists S&P doesn't disappear month to month in commodities you consume the commodity and then like it's a new commodity a month later or a few months later which like the only thing which is connecting potentially different months is storage, right? So if you have inventories in wheat, then you could connect a little bit the behavior of this month to next month. But certainly there is a lot more variety in terms of what are the risk factors that affect different months. And when we look at the front month inventories, for example, is really important aspect of what's happening. And the other thing which is funnily enough is important is actually the composition of the people playing in the market. So you would find that in the front contract of oil you will see a lot of CTAs, a lot

of market makers, a lot of retailers playing a little bit of a game in, you know...

oil futures, oil options, actually. And then as you move out to later contracts, you will find that the people who participate are the people who care about genuine production and genuine consumption. So if you go out three years out into the oil curve, it is mainly dominated by physical players who really want to hedge their production. And they want to sell it or they want to buy because they are consumers of oil. So the price dynamics at that point is about long-term supply and demand, cost of production, that sort of thing. If you look at cotton in the back of a year or wheat or any of the agricultural, then a year out we are talking about a new crop.

Okay, so the and that is actually very different from so that the front of the market is really dominated by demand because we've already produced so to speak we kind of know we know very early on throughout the year like what is going to be the crop the size of the crop. What we can surprise us is the demand shocks. As you go to further out on the curve, suddenly you're moving to the new crop. You really don't know what is going to be the supply. So you're actually more and more worried about whether there's going to be a plague, what is going to be the weather. It's going to be different risk factors. So I'm talking a lot about various risk factors. It's really exciting. But you have to ask yourself the question about what is how much diversifying risk I have. When CTA start looking at what to trade, The first question is to say, let's capture 80% of the risk with just trading the front contract. And it's very useful because the front contract is super liquid and trading costs are lowest. And lots of people are trading it and that means that there is actually good volumes to trade. It's very easy to roll the position. So it's very convenient to trade the front contract and you see that's why most of the liquidity is there. And then the question is what is relevant to this contract. I'm now trading the June contract. What is more relevant for me in terms of generating the signal? Is it the history of the June contract or is it the history of the front contracts? Okay, and these two answers are, both these answers are valid actually. And to some extent, the industry has made a choice that we are looking at essentially, most of the industry is looking at essentially the history of the first contract. And there's a good reason for that, because as you approach the front, it is all about

short-term supply and demand. It is about inventories. So the front contract, in some sense the history of the May contract last month, when it was the front contract, is actually more relevant to the June contract. And we see that, for example, as you approach expiry, the volatility of the of the June contract that maybe have been less volatile a month ago suddenly goes to match the current front contract. Sometimes it does make sense to separate them so you're talking about natural gas that's a really great example where you know summer and winter contracts are very different and the volatility is different so when you roll the position from one to another you're actually rolling potentially from one contract which is more volatile to another which is less volatile or vice versa.

And at that point actually you have a decision to make, which do you keep the volatility of the individual contract or do you keep the role contract? And it's not that we don't do it. A lot of the CTAs just do the front contract and that's actually perfectly valid. very specific contract in very specific markets when there is obviously a material difference and I think natural gas is an example. For example, we would calculate volatility on a specific contract or we would make the adjustment to when you roll the position in order to manage the risk slightly better. But that's not a, there's no real truth here. All of this, if you look at the how much risk is explained by what we do already with just with the front contract, we explained in financial universe like 95, 99% of the risk in commodities less so. So sometimes if you want to capture a different risk factor, you really have to trade further out of the curve. I think the one thing which is not trend following is other strategies like carry or trying to calculate the role yield.

in commodities, there you really have to be aware of the seasonality. Like if you're trying to calculate a carry in natural gas between the June and the December contract, good luck to you. Like the two contracts are just completely, you know, there isn't any connection between them in many ways. So understanding role yield and convenience yield and stuff like that in commodities, you really have to pay a lot more attention to the specific of each individual contract. Great stuff, Joav, and great question, Marcus, as I'm sure people can tell. I really did have the perfect guess to answer this question, so thanks for that, Joav. All right, you already alluded to it. We are going to discuss a blog post linked to a paper from our friends over at CFM

in Paris, written by two different group of people. The first one, the blog post, one of the names, that is one of the authors. It's actually Philip Sieger, who's been a guest on the podcast and the other, the actual paper, which we touched on last week with Alan. One of the authors there is Jean-Philippe Peuchot, who has also been a guest on the show. So it'll be good to dive in and see how these two things link together to some extent.

head over to you, Joav, to gently take us through these somewhat... It's not the easiest stuff to get your head around, so let's break it down. I'll do my best, but I really love this blog. And it's not trying to make any huge statement. It's really very descriptive. It talks about the evolution that we have seen in the equity markets over the last 20 years or so. Yeah, I should just mention, sorry to interrupt, I should mention that the blog post of people looking for it is called Seven Degrees of Market Structure. So just to make it clear. So it's a wonderful piece of descriptive information and it makes seven observations about things that have changed in the industry. First of all, we've seen a lot of democratization of the market and that's part of a lot of money being injected, for example, during covid into the into the retailers and the retail doesn't know like it's got lots of money and it's not it's not busy working at work. So what is it going to do?

Retail investors started flooding the equity market. And that's really quite exciting. And the other aspect that they talk about is the rise of systematic investment, because of course, you can now, if you have a broker account, which has an API, you can literally create your own systematic trading and trade from home. And that's absolutely wonderful. But the observation that I really found interesting, where they made three different observations which I think are actually related to each other and I want to speak a little bit about that. So the one thing that they make an observation is about the options explosion. We've seen a huge rise in the volumes of what's called zero day to expiry options in the index.

So normally, when I was a little lad and I first entered the financial market, you would buy, you would like S&P options were like a monthly affair, a monthly expiry affair. And then they became weekly, I think in 2012. Now they're like, you can get basically an expiry every day. And what you've seen is a huge increase in the volume of trading and funnily enough, driven

by retail. We were talking about short vol. A lot of the retail is about selling at the money straddles. So both a put and a call at the money and basically harvesting what's known as the variance risk premium. And we've seen another change in the market that we've seen over the last couple of years, which is that the market seems to look like they are becoming more idiosyncratic. And the way that we measure this, we look at what's called the dispersion index.

which actually looks at the implied correlation that we can get from how much volatility do we have in each individual stock in the index. We can add all the variants of that and then we can subtract the variants of the VIX, the index, and that number is becoming very big. So there's a lot more variance in the idiosyncratic side than there is on the VIX side, on the index side. And it's been actually a global phenomenon, not just in the S&P, but in pretty much most equity markets. If you look at the S&P sheet where they track it, they've got a dashboard tracking a correlation across multiple equity indices. Correlation is at an all-time low, i.e. idiosyncratic risk is at an all-time high. And the final observation that they show is actually the behavior of the index becoming more mean reverting, and they call it like the buy the deep psychology. Historically, buying the deep was a mistake. You bought the deep and then you were like catching the knife, the falling knife.

But over the last 10 years, buying the deep actually works much better. And if you look forward a month later, you would have made back the money. And we've seen that. We've seen that in the Trump Liberation Day. We've seen that in the equity market recently with the Homoes. So it seems to work. It seems to work. And that's really nice observations. And I kind of want to link them together a little bit because that kind of offers the mechanism to understand what's going on. So with a little bit about the options market, normally when you, a retailer or when you or I trade against and buy or sell an option in the index futures options, then we are trading against the market maker. Somebody like Morgan Stanley or JP Morgan or Goldman Sachs, they market make in the options market. And then what they do is they go out and hedge their position. So it's not like a naked position for them. They basically look at the delta of the options that they are holding and they go and how to hedge it. And in the past, when you

used to trade monthly options, okay, then, you know, today I would buy, I would trade a stock, like if I'm a retail or I'm a fund manager and I trade at 100.

Tomorrow I might be trading at 105 because the market has moved and a day later I'm trading at 120. And each time that I'm trading an option, the market maker would take the other side, then we'll put on a hedge and that hedge will change slowly because there's not a lot of gamma. Every time you need to change your hedging is when the delta of that option changes. So that's we measure it by looking at gamma, the second derivative of the price. So the gamma has got a really interesting behavior because it's very spread out when you are far from expiry. As you reach expiry, gamma gets concentrated at the money. So when you are near the strike, when the price is near the strike, then hedging becomes very expensive. As the price moves around, suddenly your delta changes a lot because the gamma is high and you need to hedge more often.

And that thing creates trading pressure. With the zero day to expiry, retail has been selling options at the money near expiry, literally at expiry. And that means that creates a very, very strong positive gamma for the market makers who are holding the position. So that's what does that mean. Unlike in the past where all the strikes were spread out and very few of the strikes were at the money near expiry, nowadays there is a concentration of gamma at the money. And that means that what happens if the price starts going up, the market maker is like long a lot of delta and they hedge it with shorting the future. They enter the market to short the future. So what does that do to the price? That reduces the price. So that pulls back the price.

back to the strike. Similarly, if the price starts dropping, the market makers will have a negative delta which they need to hedge out and they start buying the future, essentially to compensate. So to me, that's kind of anti-trend. When we think of CTAs as somebody who buys straddles, Okay, and what we have seen a flood of retail people who are selling straddles and money now You know as you pointed out selling vol is very likely to blow up, but you know that's retail for you Okay, they are just happy to collect the coupons But the net effect of that is that you having a countertrend a huge counter-trend player in the market and we understand the mechanism

for that every day every day a new Countertrend because it's a new zero-day to expiry option come into the market and is playing anti-trend game in the equity market So the fact that we saw we see out of that is The auto correlation of they in turn like in intraday returns on one day returns in entities have turned negative over time Okay, and that's really That's really reduces the overall volatility of the index, both the implied volatility and the realized volatility, because every time the price goes up, the retail is basically causing the market makers to sell back and to reduce the volatility of the index. Okay, so we've seen the index volatility to be very contained. Single stock options, single stock options or single stocks, they are a lot less affected by this process.

So when you measure the implied dispersion, which is basically the single stock variance, less the index variance, what you see is that is actually at an all-time high. So what they are describing is essentially a reflection of that process. The options explosion causing mean reversion, causing the by-to-deep psychology, and also causing the amount of risk available in the index becoming lower versus the amount of index of risk available in the single stock, right? So that's actually relating to our discussion earlier. Why might one look at single stock futures rather than index futures? And the answer is that there's now more risk available in the single stock versus the index than it ever has been because the index has been depressed.

in terms of the volatility. So that's a really, really interesting paper. And I think it's a really, really interesting observation. And I think it is related a little bit as a possible explanation to what another CFM paper, another great paper, is still your friend that you discussed with Alan. Just one question for you before we move on to that. Justin, this might be a very naive question, but if... successful in launching all these futures on single stocks. Would you expect that we would get then, because it becomes easier to trade against the momentum, so to speak, go short and all that? Would that reduce the dispersion among single stocks and actually make it look more like the index itself? No, I think that the mechanism that causes it is more the options.

If you have futures on single stocks, you probably already have the options. That's the thing. There are options already on single stocks, but there are a

lot less liquid. Even if you look at the futures, the CME is not launching futures on all 500 stocks, but only on the top 50, which are very, very liquid. So each dynamic is slightly different. We've seen growth of the option market in equities. Then we've seen a little bit in bond futures, we've seen that. So we've seen, again, moving to weekly options at different days expiry, a little bit in oil and stuff like that. Basically, the short term implication to trend following is actually quite interesting. But it's not necessarily, I think that's not going to be the driver. I think what's driving the single stock futures is more to do with the fact that retails want to short. I think everybody is kind of calling the top of the AI.

not the top of the market but the top of the valuation from a valuation perspective. The S&P is very highly valued at the moment. And that gives a very easy mechanism for people to be able to do too short. And that's I think where it's coming from. But I don't think that's what driving doesn't affect CTAs that much. All right. Well, you already mentioned it. And we're heading back to the paper Alan and I just touched on last week, which is the a microstructural account of the demise of short term trend following again from the good people over at CFM. So tell me, well, maybe I should make one thing clear and maybe maybe you can clear it up completely. I think last week and I probably did not spend enough time on the paper, but I think last week I might have left the impression that it was kind of the tick size that had changed somehow and that made it or that impacted the the short-term trend following so to speak but actually it's the volatility normalized tick size and that is very different from just the tick size so maybe you can just set the record straight and just explain what they mean.

when they say the normalized tick size, volatility normalized tick size. Yeah, but it's exactly what it says on the team. So when the CME or any other exchange sets up the market, they would set the minimum size that the price can move in. And it's normally set up in the way that the market trades at the time. And it's reasonable for the size of their market. But events happen. If I look, when you set up a certain tick size for an equity market, that might have been 30 years ago, the market has risen in value. Unlike bond futures, for example, which I love and trade, the price stays roughly at 100. Give or take, it might go to 110, 120, but the TY is trading at the moment at 108.

the market is that now the tick size is that is then the volatility just a tick size is basically the size of the tick divided by the volatility of the underlying price. Now inequities you set up the market 30 years ago the market has gone up you know we've seen the equity markets going up you know hundreds of percent. And now when I look at the volatility, suppose the volatility of the index is still 20%, but it's 20% of a much bigger number. So, you know, the S&P has changed from about a thousand in 1995 to about seven. Like it's gone up enormously, right? So the tick size in the equity space has become much smaller. So if I look at the CFM paper, I've actually done the maths. So you go back and there's a very large, like the proportion of equity market in the small tick size is much higher than the proportional market in the large tick size. Now, they do have a little bit of, I think at the end of the appendix, appendix F, at the end of the paper, they look at within each asset class what is the variation. The picture there is actually a lot more nuanced, I would say.

But I think part of the result is essentially the effect that we're seeing here, which is to do with equity markets not performing well. Now the question why equity markets have not performed well, you can take the, they provide an explanation about the sort of the sparsity of the order book. I think what this paper, this blog provides an alternative or a complementary explanation, which is to do with the nature of the players that have entered the market and they are very much playing the anti-trend position. And it's kind of nice because here we understand the mechanism. It's much clearer to see what the mechanism is that is driving, which is playing counter-trend. All right. So before we leave the papers and the blog post, maybe just talk a little bit about the relevance for trend and maybe specifically for short-term trend following of all of this.

Oh, absolutely. So I think it's important whenever you do a bit of maths is to have a little bit of back of the envelope understanding of how much does that impact the trend. So the impact that we're talking about is about reduction in say one day auto correlation of equities. And we can think about what does a 1% negative auto correlation mean for trend following. So if you were just trading If you're a trend follower, which takes yesterday's price and uses it for today, then that will be a 16 basis point sharp, negative 16 basis point sharp on your performance. 1% of correlation

times 16, because it's square root of 2, the number of days in the year, you get hit 16 basis point sharp. If you're a fast trend follower, even if you're within our industry, We are trading like the example they gave was a 520. The amount of weight that you have on that short term effect is something of the order of the magnitude, one in five, if you're trading a 520. So maybe 20% of that number. So your single market drop in sharp to 1% drop in auto correlation is going to cost you something like three basis points of sharp.

And that's not that much actually. It magnifies itself if I look at the equity portfolio as whole. So there is an effect here, which is more general. So if I do that times like you, some diversification in the equity market, you don't have that much of a diversification. So maybe we multiply by two. So that costs you six basis points of 0.06 sharp for short-term trading. That's not that significant actually and that's one of the reasons why I think that trends are driven really by macroeconomic events more than just the order book structure. Of course the order book structure and the intraday behavior becomes very important as you move faster and faster, but even for like fairly fast short-term traders. This is not a 1% is not that significant. Now we have seen maybe a drop of 3% in auto correlation in the equity market over the last 20 years. And that's kind of becoming painful, right? It's becoming a little bit painful. So we said it's about 20 basis point sharp for short-term traders. For long-term traders, the slower you are, you're basically staying away out of the game. So it's a lot less the alpha is driven a lot more by long-term trend, like secular trends that you observe in the market and macro events and so forth, a lot less than sort of the intraday. And that's, I think, one of the reasons why the industry has drifted to slightly so, because you're actually harvesting a different source of like, you're not playing in the same market as those retail traders.

I think another way that I'm thinking about these things is just from memory, the conversation we had with Toby Crabill not too long ago where he talked about that. He had noticed that, for example, markets that generally traded higher at the open in the old days, and I'm not going to quote the correct numbers, but it's going to be fairly close. But I think he said in the old days, market that opened higher, they would generally also move higher and close higher during the day, something like 58% of the time. Now, in the last few years, we're down to 54, 53% of the time. That

doesn't sound like a lot. He said it actually makes a big difference for if you're a short-term manager, whether you get that follow through.

No, absolutely. If you're playing in that short-term space, this is brutal. It's a headwind and on top of execution costs, it's painful. But as you move to the two-month horizon that CTAs are trading, now the amount of risk that you have on that one-day observation is a lot lower. Let's change direction a little bit here. The next paper that you brought along is called Thematic Investing as Missing Factors by Wei Li. Again, I think a lot of the papers that we touch on here in the podcast for those who are not on my Sunday email list, that's probably where I tend to share them. I'm not so good and consistent in terms of sharing them in the actual show notes. But anyways, you brought this along. And yeah, very interested to hear how that ties into today's conversation. Absolutely. So first of all, a shout out to Robert Kuzovsky and the Imperial College Conference on Quantitative Investing. It's a wonderful conference that has been running for 18, well, more than 18 years, but this is the 18th conference.

And it's organized by Nick Balthas as well. So a shout out to Goldman Sachs and for supporting it as well. And it's a wonderful conference. They choose really good papers. And I was listening to that paper and I thought this is really relevant to the equity discussion. Because when we talked about the dispersion, we talked about what's, you know, A lot of the risk is no longer in the index. But we have to understand that the question is, where's the risk? What is driving equity returns, single stock equity returns, if it's not the index? So in the good old days, the index might have explained 35%. These days, it might be explaining 30% or 25% of the index and then of the total variance. And now the question is, what happens to the remaining variance?

And you know, quants have already been, you know, kind of asking this question for many, many years. And of course, industry is another one, industry specific. So it might not be, you know, a health industry right now during the World Cup. It might be the entertainment business industry in the US. No, no, no, it's a real, it's a real, there's industry explains it a little bit. And then of course, in the 1970s, there was a Rosenberg guy who I can't remember his first name, and he came up with the Barra factors. That's

another amazing innovation that it was like a risk model. It says, well, I've noticed that value stocks move together and growth stocks move together. There's a certain factor like leverage or growth or value that also explains the co-movement of stocks.

And what I find really interesting about the current situation, there's an evolution of thematic investing in the industry. And themes are another way of explaining the volatility, right? So we said the index may be explaining, say, 30%. When we add industry-specific volatility and we add barra factor industry volatility, we get up to maybe 50% of the volatility. And I think the paper and the presentation by Wiley in the conference was amazing, really, really good. And he's saying, well, okay, so we've explained 50% of the volatility, we understand. What's happened to the other 50%? Themes. What are themes? Themes are like risk factors, but the things that are transient and they are driven by news. So, World Cup is a theme, okay? Or, you know, straight-of-home moves.

it's a theme. COVID, these are like very big themes, but there might be small themes. So it might be single stock semiconductors. Semiconductors is a great example because it's both a theme and also an industry sector. So I think what the paper says, well, okay, how much extra volatility I can assign to semiconductors, which is not explained by just the semiconductor industry moving together, okay? How much can I harvest that? And they've done a lot of work in trying to sort of basically first of all mapping news and mapping sort of events that happen, right? The semiconductors is playing at the moment because there's a huge demand for obviously for GPUs. And he says, well, There are two observations that he wants to make the first one is that the total amount of risk that we can explain is another 5% So we have explained 50% as 5% there's still 45% of risk that we like which is genuinely idiosyncratic, right? It's maybe we can't explain at the moment or like there's no Which which is fine like we have idiosyncratic behaviors of companies. They are just idiosyncratic by nature But then is but the other observation he says that if investors are interested in it, okay How do you harvest it? Okay. And I think his paper is really interesting is that we've seen a growth in the ETF industry, right? So if you go to Morningstar and say, I want to buy, you know, let the military industry something, because it's there's a war going on.

Then there is an ETF for that. In fact, you get very specific ETFs for like health care and health care manufacturing and health care heart surgeons and health care kidney, right? You can get ETFs for almost everything and like there's thousands of ETFs complaining it. The second one is of course the QIS. you speak to Morgan Stanley, you speak to Goldman Sachs, you speak to these guys, they will say we can create QIS baskets harvesting, World Cup harvesting, Iran war. But what he says is actually sometimes these do not actually capture the volatility. The paper is talking about capturing additional orthogonal volatility. And he said he observed that a lot of ETFs are actually too much exposed to basically to the industry, or to the currency, or to the country, or to the general barra factors that we've seen and are actually not representing. So if an investor wants to have more pure, theme-based exposure, he's actually better off essentially constructing this on his own or her own for that matter. So it's a really nice paper because it's less about prediction.

it's really about risk, about trying to identify where is the risk coming from. And I think when we design, I think it goes back to the question we asked in the commodities, is like how much extra risk do you have in going to single stock futures? How much extra risk do you have in going to the third or the fourth contract on the commodity curve? All of these, this is much more important to have sorted out before you even try to predict things. So I really, really, really like this paper because it's like such a nice framework to understand, okay, what is thematic investing, which is all the craze at the moment? How much risk is there in that? And part of the whole thing about themes, obviously a lot of people will know the AI theme at the moment, but it's narratives at the end of the day. And funnily enough, there's another paper that you brought along by the title of narrative momentum. So I thought we'd probably spend a little bit of time on that before we wrap up completely. But this is a paper by, I think, Ronny.

Ronny Sartka, yeah. So Ronny Sartka, an amazing academic quant, is done a lot of work on seasonality. And this paper is about a really unique data set that they've done. They've curated news and like millions of articles. 13.3 million articles. Yes. So, I mean, it's a Sisyphean work between 11, 12 years. And what it does give them, which is really nice, is a real point in time of the evolution of stories. Because nowadays, if you run a search about what

was the news in 2012, good luck finding out. The internet does not remember. But they do. So it's really nice that they have a really good point-in-time data set to say, OK, can we understand how themes and narratives evolve? And what does that imply to prices? And I'm going to cut a very long story short. What they do is they look at their themes, which are like war, which investors tend to react instantly, which is not surprising.

So, but a lot of stuff is like it just evolves very very slowly. So he has a lot of themes that take months to evolve. In fact, he looks at like six months horizons. And what he finds is there is the underreaction like it takes time for. For investors and we are very familiar with that we are strength followers right away indeed exactly and in many ways like the whole point of that is that in the absence of complete information in the market the fact that the price has gone up or the fact that other people are talking about a theme. Actually makes you evaluate your own valuation in the absence of complete information right so one of the reasons.

the way that we form our own valuation of what a commodity is worth or what an equity is worth is through essentially seeing what other people are talking about or buying, right, or the way that price action behaves. So it's a really, it's perfectly understandable and I would say explainable and I would say legit rationale. And he spends a lot of time then constructing exactly like the first paper, sort of a beta for each of the stocks to the news. And then he shows that that indeed actually does make money. You can essentially trend follow the news. And as the news collapse, then you kind of essentially move away from those stocks. And you make money that way as well.

I really liked it. He looked at the back test. He looked at the correlation to general momentum, momentum that we cannot know in life, because of course, the first question is how related it is to price momentum, and it shows that it is unique. I think there's a mathematical issue I have with the paper, and there is a reality check on the paper, I think. The mathematical check is actually to do with the calculation of the beta of the stock to a narrative and that's actually something which is difficult for us, right? So if I'm trading the first contract or the second contract or the third contract in commodities, we do have a good history to understand what is the

correlation between summer natural gas and winter natural gas. There is, we can do some maths over a long period of time, but to understand what the beta or the correlation between a certain stock and a narrative, it's a much more difficult exercise, especially as we've already established that only about 5% of the volatility is driven by the narrative. So that makes the estimation error on being able to construct a genuine basket difficult.

it's not a it's a very subtle process and you know it's mathematically possible to diagonalize matrices but whether what we get is correct is not i mean i think i've spoken to one of the big uh one of the big banks and i think they they do a qas product and in that case actually i actually like the way that they do an uh a human overlay of the basket for the themes that they're doing so they're looking at the they're saying well this stock it might not show in the price but we we kind of understand why Coca-Cola will benefit from the World Cup, for example. And I think I like that approach. Of course, completely non-quantitative makes it very difficult for us to trade. But from a mathematical point of view, I think that's the weakest part of the paper. But I think I have a much more generic issue, which is that the way we consume news has changed.

So the paper is definitely correct for the universe between 2012 to 2023, if I remember correctly. That's the period that it was covering. But with the rise of AI, with the rise of our ability to harvest text and news. And I mean, you see that in the market, you see that analysts these days, machine learning will basically churn all of the news. on the stocks that are interested and will present it like instantly. It's just insane that ability. And this is the human analyst, right? The CTAs, the systematic funds, they do that already. It's a done problem now. over the last five years has become a very much solved problem. So I think that's the, that underreaction over reaction is potentially going to be up the way quite quickly. So that's, that's my issue with that market. So, yeah. So before we move to the last kind of small point where, where we wanted maybe to, to, um, sum it all up and, uh, um, and talk about kind of different ways you can trend follow equities. Um, it, when you, when, when you just said that, uh, before, It actually reminded me of a section of this interview I mentioned in the beginning with Jason Wong, where he talked about that with agentic AI as it comes, news is

actually not anymore being pitched from some kind of database. It's actually being generated for each of us individually.

So what is the news? Well, it depends on who you ask, so to speak, as far as if I understood him correctly, which is kind of frightening and intriguing. No, it's a great observation. It's actually something I haven't thought about, which is that we all live in our own bubble, like the social network basically presents us with the news that we like to think about. And that creates our own bias. in the trading strategy, right? So you're right. In that sense, the truth, the narrative truth is almost individual. Yeah, yeah, yeah. Anyways, just to sum it all up, before we wrap up, so to speak. So how do you create Adversified? Okay, so like we've covered like a huge range. Obviously, us as CTAs, we trade equity.

you know, equity index futures that we understand now covers about 30% of the equity risk that is available to harvest. And we've talked about sort of short term facing challenges, but medium term and long term equity trend actually is all nice and well. Okay. And then we think about industry and industry is actually reasonably easy to trade. There's lots of plenty of ETFs which offer it. There are a lot of actually futures on specific sectors, and that will give you a nice diversification. You have to be aware that you don't want to double count, right? So at the moment, trading the semiconductors and the S&B, you're basically trading the same thing. So I think construction, you have to start thinking about construction. And then we move on to the borough factors, right? At that point, you really have to think about the construction, you have to think about shorting, you have to think about long, like you're constructing it from individual stocks, but it's difficult, right? And it requires you to normalize to get a truly diversified portfolio. And it's, and it doesn't, there's less and less risk. And of course, narrative is even worse in that respect is because at any time it might be covering 5%, but you're your risk factor changes over time. So you're not just catching up to trend, you're also catching up to the risk factor, which is trading right now. And that's really difficult. But I think what is nice is that you can actually create sort of a CTA, a trend program in equities, which is like almost uncorrelated to the first component.

So all of these things, the aim for that is to create and I don't think I'll be telling huge state secrets that some of the companies that I've worked in the past have been doing that. So I'm describing something in the process that a lot of people have already gone through and that diversifies and that's a good thing. But equity is a really rich universe that you can create a diversified trend portfolio in it. Yeah, well, we certainly made some managers happy to hear that strength following on single stocks is an interesting and potentially profitable venture, whilst most CTAs probably still stick to the index for their purposes in terms of trigon exposure. Anyways, this was super. You're very educational. I really appreciate that.

And to everyone listening today, if you want to show some appreciation for the preparation, which is not insignificant for these conversations, then do go to your favorite podcast platform and show your appreciation for the co-host to come every week and share some of this really insightful information. Next week I'll be joined by another great co-host and that is Nick Baltus that we just talked about earlier today and no doubt that's going to be timely, fun, insightful. And if you have a question for Nick, as always you can email them to info.toptraderonblock.com and I will do my very best to bring it forward. Now you are the World Cup final. won't have one of our teams. Denmark didn't even make the World Cup. Switzerland is out. So is England. England, yeah. But it might still be a good game. So I think on that note, we have something to look forward to other than the release of this episode this weekend. Looking forward to Spain winning. Looking forward, you're looking forward to Spain winning. I look forward to the best team winning, whoever that might be. So anyways, Joav, thank you so much. So from Joav and me.

Thanks ever so much for listening. We look forward to being back with you next week. And in the meantime, as always, take care of yourself and take care of each other. We have some amazing guests lined up for you and to ensure our show continues to grow, please leave us an honest rating and review. It only takes a minute and it's the best way to show us you love the podcast. We'll see you next time on Top Traders Unplugged. This podcast expresses the views of its hosts and the guests appearing on the podcast as of the data bits recording and such views are subject to change without notice. Top Traders Unplugged do not have any duty or obligation to update

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